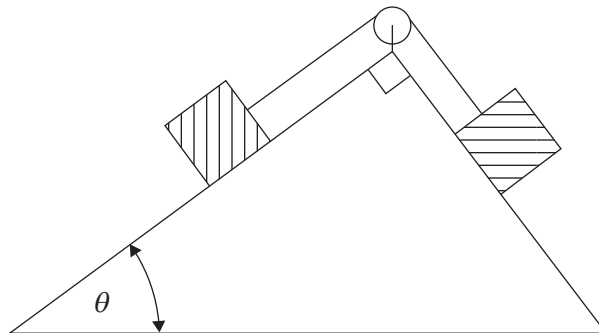


PAPER 1 / QUESTION 51

25 April 2024 / TZ2 (file label) / HL Paper 1 / original question 4 / 2224-6707

51. Two blocks of equal mass are connected by a light string that passes over a frictionless pulley. The blocks slide at a constant velocity on inclined planes that are at right angles to each other. One of the inclined planes makes an angle θ to the horizontal such that $\theta < 45^\circ$.



Which of the following statements is correct?

- A. The net force acting on each block is the same.
- B. The weight force acting on each block is different.
- C. The magnitude of the normal force acting on each block is the same.
- D. The magnitude of the force exerted by the string on each block is different.